Assessing VLMs using High-FPS Gameplay

Anonymous Author(s)
Affiliation
Address
email

Abstract

Vision Language Models (VLMs) are typically benchmarked on static images or video clips, leaving their ability to perceive dynamics and act in real time under-explored. We introduce AtariBench, a benchmark that evaluates spatiotemporal understanding and sequential decision-making by requiring VLMs to operate directly on high–frames-per-second Atari 2600 gameplay across 10 diverse games. Experiments show consistent shortcomings: models fail to reliably detect object motion and direction, exhibit brittle step-to-step decision-making, and often over-rely on textual history instead of grounding actions in the visual stream. By isolating these core perception-to-action skills in a controlled, scalable setting, AtariBench surfaces concrete directions for developing VLMs that reason and act effectively in dynamic environments.

1 Introduction

A significant limitation of current Vision-Language Models (VLMs) is their evaluation on static scenes, which hinders their ability to operate in dynamic environments [20, 16, 4, 10]. To bridge this gap, models must develop three fundamental capabilities: 1) Spatial-Temporal Understanding, the ability to grasp intuitive physics like time, distance and speed; 2) Sequential Decision-Making, the capacity to process a stream of information to determine the next action; and 3) serving as Action-Oriented Agents, models that can perform tasks, not just describe them.

To evaluate these interactive capabilities, we propose AtariBench, which evaluates VLMs on 12 Atari 2600 games. The Atari environment [2] has long served as a canonical testbed for agent-based AI [18, 1, 7]; here, we repurpose this classic platform to specifically evaluate modern VLMs. In our setting, models are tasked with making sequential decisions directly from high-frames-per-second (FPS) gameplay streams, requiring them to ground their actions in visual dynamics rather than static inputs. Our approach thus marks a clear departure from static benchmarks [25, 22]. Furthermore, while other vision-to-action benchmarks exist, their strategic and visual complexity can obscure the foundational reasoning failures we aim to isolate [8, 31], or on image-based planning rather than video dynamics [30]. AtariBench , in contrast, uses simple, constrained physics as controlled environments to specifically measure these foundational abilities within a scalable environment suited for rapid iteration.

Our experiments with AtariBench reveal a cascade of critical failures. We observe that models consistently struggle to detect object movement and direction from the visual input. This fundamental perceptual error leads directly to poor in-game choices, as the model cannot formulate an appropriate plan. Most significantly, we show that the models fail to ground their actions in the present visual context [5], often ignoring the screen and instead attending more to text from the prompt history.

The AtariBench framework offers several practical advantages for research. It provides scalable environments for rapid iteration, a key advantage for model development. Its simple yet diverse game rules enable evaluation across a wide range of tasks [17, 9]. The framework also allows for controlled

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difficulty, where game mechanics can be isolated to probe specific capabilities, and provides well-defined evaluation metrics through in-game scoring, offering an objective and quantifiable metric of task performance.

The insights from our benchmark have profound implications for VLM development. AtariBench makes a significant leap by shifting evaluation from passive description to active agency. This clearly points to future research directions: building models with a more robust grasp of intuitive physics, improving their ability to make decisions in dynamic settings, and ultimately, creating the truly interactive agents needed for future applications.

2 Related Work

Game-Based Benchmarks for Language Models. Recent years have seen a surge of work on game-based benchmarks from diverse genres, including grid-based puzzles [19], simulation and resource-management strategy such as Factorio [11], turn-based role-playing with strategic combat such as Pokémon [15], and narrative reasoning tasks such as murder-mystery games [29]. Others evaluate natural language reasoning through text-based or conversational games [6, 14, 23, 13], lacking visual understanding. Multimodal gaming benchmark like BALROG [21], LMAct [24], and LMGame-Bench [12] assess planning using image inputs, but their tasks are dominated by grid-based games that do not directly probe fine-grained spatial–temporal reasoning (e.g., movement trajectories, speed, or continuous dynamics) as found in Atari. VideoGameBench [31] evaluated VLMs on AAA games, but most models make little progress due to the extreme complexity of these games, limiting its ability to distinguish between models. Cradle [26] builds computer-use agents for such games but relies on modules like OCR and object detectors, making it hard to isolate VLM reasoning. In contrast, Atari games provide a lightweight testbed with simple rules and visions (VLMs can make progress), sufficient to probe movement and timing. Atari-GPT [28] explores the use of VLMs on Atari environments, demonstrating initial feasibility. Our work, however, significantly extends this by constructing a systematic and reproducible benchmark designed for rigorous evaluation of VLM spatial–temporal reasoning. This involves a broader set of games, standardized settings (e.g., FPS, context length, instruction format), and repeated trials to ensure statistical reliability, thereby moving beyond a proof-of-concept to establish a robust framework.

Video Understanding, Action Recognition and Decision Making. Significant research has been dedicated to video comprehension and action recognition, driving the creation of a wide array of benchmarks. General-purpose benchmarks such as MVBench [16], Video-Bench [20] and Perception Test [22] contain tasks that require models to demonstrate basic world understanding, capture temporal dependencies, and recognize actions. Beyond these broader assessments, there are benchmarks specifically emphasizing fine-grained action recognition—for instance, ActionAtlas [25] assesses whether models can distinguish subtle differences between human motions, often requiring domain-specific sports terminology, and MotionBench [10] focus on motion comprehension through video-based QA. These video–action benchmarks test perception and temporal reasoning, but they typically focus on classifying or describing actions rather than taking them in dynamic environments. Recent work has introduced action-taking benchmarks on VLMs: EmbodiedBench [30], which emphasizes image-based planning for embodied agents, and PCA-Bench [4], which uses multiple-choice action prediction across diverse domains, although both fall short of testing spatial–temporal video dynamics in interactive environments.

3 Method

Our research introduces AtariBench, a novel framework for evaluating the physical reasoning and agency capabilities of Vision-Language Models (VLMs) through high-FPS gameplay in Atari 2600 environments. This section details the philosophy of game choice, the design of rewards and termination, and the xx of our gaming agent.

3.1 Setup

We have developed an evaluation framework using the Arcade Learning Environment (ALE) [2], a standard testbed for AI agents featuring a diverse set of Atari 2600 games. Our framework interfaces

with the environment through the Gymnasium library [27], the successor to OpenAI Gym [3], which provides a standard API to communicate between the running algorithms and environments.

The core interaction with the ALE is frame-based, governed by the function `env.step(action_id)`, which returns an observation (`obs`) as an image frame, along with reward and state information. Our evaluation environment is configured to present a consistent and dynamic challenge to the agent by adjusting difficulty and number of game episodes.

Game states are rendered as RGB pixel arrays at the native resolution of 210×160 pixels. ALE runs at a native rate of 30 FPS. In our evaluation, we downsample the visual stream to 10 FPS, and the agent decides one action for each of these frames by setting frame_skip=3. Thus, the visual update rate and the control loop are synchronized at 10 Hz, ensuring that every displayed frame corresponds to an action opportunity.

To create a consistent and manageable challenge, we make several further adjustments:

- **Game Difficulty:** Each gameplay episode is configured to last approximately 25 to 50 seconds, ensuring a standardized duration for evaluation. This is achieved by adjusting parameters such as the number of lives, number of episodes, or by enforcing a number of frames per episode, depending on the game's mechanics.
- **Reward Normalization:** To simplify the learning signal and the prompting engineering process to describe game rules, we normalize game rewards. Positive events that contribute to the game's objective are mapped to +1.0, negative events to -1.0, and neutral events to 0.
- **Stochasticity:** To remove environmental stochasticity, we set the probability of action repeat to be 0 ("action_repeat_probability=0"). This ensures that actions are executed deterministically and are not stochastically repeated from the previous timestep.

3.2 Game Choice

Our goal is to probe intuition-physics and closed-loop control under fast visual dynamics, so we select games that (i) require continuous motion perception (objects move on screen), (ii) demand fine-grained timing or aiming, and (iii) yield non-zero rewards within short episodes. As shown in Table 1, our game selection varies systematically along multiple capability axes beyond genre, including primary skill, action space, movement axes, reward density, visual complexity, and tempo.

3.3 LLM-Agent Interaction Protocol

Observation and Prompt Formulation. At each decision point, a structured prompt is constructed and sent to the LLM, composed of several key components:

- Game Instructions and Rules: A text description outlining the game's objective, rules, and scoring system.
- Action Space Definition: A clear mapping of textual commands to the available in-game actions (e.g., "move left", "fire").
- Historical Context: A history of recent game-play clips, including the visual frames, actions taken, and rewards received. This provides the model with temporal context about the immediate past.
- In-Context Reward Examples: To guide the model towards successful strategies, the prompt includes examples of past clips where the agent achieved a positive reward. This helps the LLM to associate specific actions and states with desirable outcomes.
- Current Game State: The most recent visual frame from the game is provided, showing the agent its current situation.

Both the historical context and in-context reward examples are constructed from clips. A clip is defined as a sequence of visual frames that captures the environment's state transition during the execution of actions generated from a single LLM response. This allows the model to perceive the immediate consequence of its collective actions.

Table 1: Atari game selection across genres and skills. Move: Action: H=horizontal, V=vertical, B=both, A=fire/attack/throw. Terminate means the termination strategy we used. Reward density and visual load are qualitative. Tempo reflects timing precision required to play this game.

Game	**Primary Skill**	**#Obj**	**Action**	**Terminate**	**Reward density**	**Visual load**	**Tempo**
			Shooter				
AirRaid	Aiming & Avoidance	2	H+A	Time	Dense	Low	High
Assault	Aiming & Avoidance	3	H+A	Lives	Dense	Low	High
BeamRider	Aiming & Avoidance	2	H+A	Lives	Dense	High	High
DemonAttack	Aiming & Avoidance	2	H+A	Lives	Dense	Low	High
NameThisGame	Precision Interception	4	H+A	Time	Dense	Low	High
Phoenix	Aiming & Avoidance	2	H+A	Lives	Dense	Low	High
RiverRaid	Scrolling Navigation	3	H+A	Lives	Dense	High	High
Seaquest	Multi-task Navigation	3	B+A	Lives	Dense	High	High
TimePilot	360 degree Navigation	2	B+A	Time	Sparse	High	High
Robotank	first-person Navigation	2	B+A	Time	Dense	High	High
			Sports				
Boxing	Rhythmic Combat	2	B+A	Time	Dense	Low	High
IceHockey	Positional Strategy	1	B+A	Rewards	Sparse	Low	Med/High
FishingDerby	Depth Targeting	1	B+A	Time	Dense	High	High
Tennis	Predictive Timing	1	B	Rewards	Sparse	Low	Med/High
			Action				
Breakout	Rebound Control	1	H	Lives	Dense	Low	Med/High
Freeway	Gap Identification	2	V	Time	Sparse	High	High
Gopher	Reactive Pathing	2	H+A	Lives	Dense	High	High
JourneyEscape	Scrolling Navigation	1	B	Time	Dense	High	High
			Maze				
PacMan	Route Planning	2	B	Lives	Dense	High	High
Qbert	Route Planning	1	B	Lives	Dense	High	Low

Representing Game States. We encode each game state as a short sequence of images rather than as raw video. This choice is motivated by two factors. Firstly, most current LLMs cannot directly process video input. Second, for models that do support video input (e.g., Gemini 2.5 Flash and Pro), we observed no performance improvement compared to providing an equivalent sequence of images. To make the prompt structure explicit, we explicitly mark the start and end positions of each component (recent history, reward, and current state).

Action Generation and Execution. The LLM processes this multimodal prompt (containing both text and images) and generates a response. This response is structured to ask model to generate a "thought" process, followed by a specific sequence of actions to be executed. Our framework parses this textual response, converting the commands into a series of action IDs that are then sent to ALE. These actions are executed sequentially until the next decision point for the LLM is reached.

3.4 Agentic Harnesses

3.4.1 Memory

longer recent history

3.4.2 Reflection

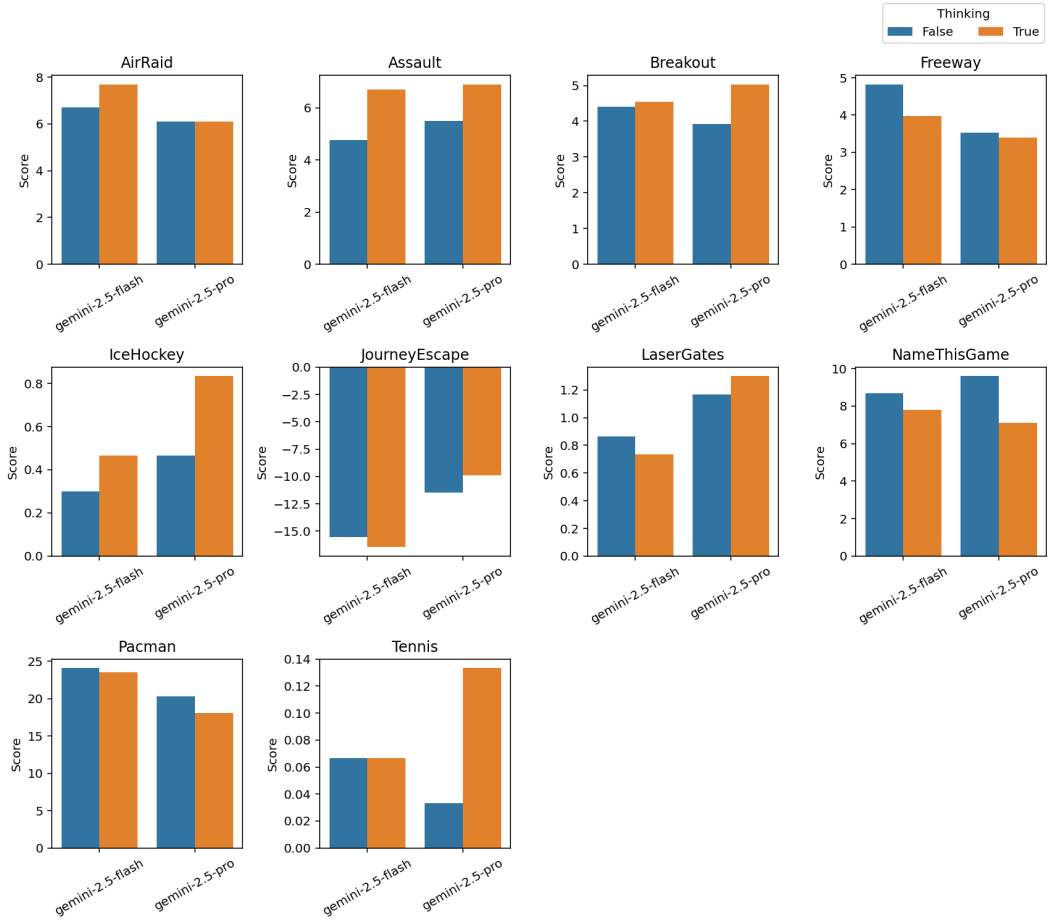
textual input non-zero reward history

4 Experiments

4.1 Results

In this section, we present the evaluation results for four distinct large language models: GPT-5, Gemini 2.5 Pro, Gemini 2.5 Flash, and Claude Sonnet 4-5. To ensure the statistical robustness of our findings, each experimental setting was rigorously repeated for 30 independent runs. See detailed results in Figure 1. We found that Gemini 2.5 Pro with default thinking outperformed other models by winning 6 out of 10 evaluated games, demonstrating a notable advantage in Atari games.

Figure 1: Model Performance Across Atari Games.



4.2 Correlation Analysis

Our correlation analysis was performed on a dataset comprising model performance metrics across various Atari 2600 games. We analyzed the averaged reward achieved by various models considering both with and without thinking mode. Each game's performances on these models are represented using a vector, and we compute the pairwise Pearson correlation among these vectors and show results in Figure **??**.

This correlation analysis revealed significant inter-game dependencies (when greater than 0.8). Breakout, Tennis, and Ice Hockey exhibited strong correlations, likely due to their shared demand for tracking small, fast-moving objects. PacMan and Freeway also showed high correlation A key finding across all cases was the superior performance of models without the "thinking" component. Additionally, JourneyEscape and LaserGates correlated significantly, attributable to their common

obstacle avoidance mechanics. These insights highlight shared skill requirements and performance trends across the analyzed Atari games.

4.3 Qualitative analysis

This section showcases representative failure cases across the evaluated games and offers qualitative insights that point to clear directions for improving the models.

Spatial-Temporal Understanding. As discussed in Section 4.4, models can correctly identify salient objects and their relative spatial relationships, yet they often fail to reason about motion. In Breakout, for instance (Fig. **??**), the model infers a movement direction that contradicts the ground truth. This weakness in spatiotemporal understanding limits performance on broader video-understanding tasks that require accurate motion perception, such as activity recognition, object trajectory tracking, and event understanding in dynamic scenes.

Reflection on Failures.

Long context.

Reasoning.

4.4 Error analysis

To diagnose potential causes for the models' suboptimal performance in Atari games, we conducted a targeted evaluation. This diagnostic approach involved presenting the VLMs with simple, focused questions about specific game states, for which the ground truth answers were manually labeled. This allowed us to pinpoint the failures from critical capabilities relevant to gameplay:

1. Perception: To test fundamental object recognition and spatial awareness, we posed the question: "Q1: Is the ball to the left or to the right of the screen?" The ground truth for this query was manually labeled based on location of the ball, providing a clear assessment of basic visual comprehension.
2. Motion Comprehension: To evaluate the model's understanding of dynamic elements and their trajectories, we asked: "Q2: Which direction is the ball moving? Choose from (up, left), (up, right), (down, left), (down, right)." The correct direction was also manually labelled by observing the ball's movement across a short sequence of frames, allowing an objective measure of motion inference.
3. Decision Making: To probe the model's ability to infer optimal actions based on the immediate game state and rules, we asked: "Q3: What is the best next action for the paddle to catch the ball? Choose from left or right." The ground truth for the optimal action was manually assigned based on a simple heuristic (e.g., moving the paddle towards the ball's projected horizontal position), enabling us to assess the model's decision-making logic.

Table 2: Failure case analysis results.

Model	Q1: Recognition	Q2: Motion	Q3: Decision
Random Guess	0.50	0.25	0.50
Gemini-2.5-flash	0.90	0.42	0.58
Gemini-2.5-pro	**1.00**	**0.58**	**0.72**

The results of this evaluation are presented in the Table 2. This shows that while VLMs exhibit strong perceptual abilities (Q1: Recognition), their performance significantly degrades when assessing object motion (Q2: Motion) and making spatial decisions (Q3: Decision). This suggests suboptimal performance in video games is primarily due to limitations in dynamic scene comprehension and sequential decision-making, rather than basic perception.

5 Conclusion

In this work, we assessed the physical reasoning and decision-making capabilities of modern VLMs by employing a high-FPS gameplay benchmark built on the Arcade Learning Environment. Our findings reveal that even state-of-the-art VLMs struggle with the intuitive physics required for dynamic, interactive tasks. While providing a longer context history generally improves performance, a targeted analysis pinpointed the models' primary weaknesses not in basic perception, but in comprehending object motion and spatial dynamics. Our results suggest that current VLMs function as passive observers rather than active agents, and that future progress in developing physically-grounded, embodied AI will necessitate training directly within these kinds of interactive, real-time environments.

References

[1] Adrià Puigdomènech Badia, Bilal Piot, Steven Kapturowski, Pablo Sprechmann, Alex Vitvitskyi, Zhaohan Daniel Guo, and Charles Blundell. Agent57: Outperforming the atari human benchmark. In *International conference on machine learning*, pages 507–517. PMLR, 2020.

[2] M. G. Bellemare, Y. Naddaf, J. Veness, and M. Bowling. The arcade learning environment: An evaluation platform for general agents. *Journal of Artificial Intelligence Research*, 47:253–279, jun 2013.

[3] Greg Brockman, Vicki Cheung, Ludwig Pettersson, Jonas Schneider, John Schulman, Jie Tang, and Wojciech Zaremba. Openai gym. *arXiv preprint arXiv:1606.01540*, 2016.

[4] Liang Chen, Yichi Zhang, Shuhuai Ren, Haozhe Zhao, Zefan Cai, Yuchi Wang, Peiyi Wang, Xiangdi Meng, Tianyu Liu, and Baobao Chang. Pca-bench: Evaluating multimodal large language models in perception-cognition-action chain. In *Findings of the Association for Computational Linguistics ACL 2024*, pages 1086–1104, 2024.

[5] Shiqi Chen, Tongyao Zhu, Ruochen Zhou, Jinghan Zhang, Siyang Gao, Juan Carlos Niebles, Mor Geva, Junxian He, Jiajun Wu, and Manling Li. Why is spatial reasoning hard for vlms? an attention mechanism perspective on focus areas. In *Forty-second International Conference on Machine Learning*, 2025.

[6] Anthony Costarelli, Mat Allen, Roman Hauksson, Grace Sodunke, Suhas Hariharan, Carlson Cheng, Wenjie Li, Joshua M Clymer, and Arjun Yadav. Gamebench: Evaluating strategic reasoning abilities of llm agents. In *Language Gamification-NeurIPS 2024 Workshop*, 2024.

[7] Adrien Ecoffet, Joost Huizinga, Joel Lehman, Kenneth O Stanley, and Jeff Clune. First return, then explore. *Nature*, 590(7847):580–586, 2021.

[8] Linxi Fan, Guanzhi Wang, Yunfan Jiang, Ajay Mandlekar, Yuncong Yang, Haoyi Zhu, Andrew Tang, De-An Huang, Yuke Zhu, and Anima Anandkumar. Minedojo: Building open-ended embodied agents with internet-scale knowledge. *Advances in Neural Information Processing Systems*, 35:18343–18362, 2022.

[9] Jesse Farebrother and Pablo Samuel Castro. Cale: Continuous arcade learning environment. *Advances in Neural Information Processing Systems*, 37:134927–134946, 2024.

[10] Wenyi Hong, Yean Cheng, Zhuoyi Yang, Weihan Wang, Lefan Wang, Xiaotao Gu, Shiyu Huang, Yuxiao Dong, and Jie Tang. Motionbench: Benchmarking and improving fine-grained video motion understanding for vision language models. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 8450–8460, 2025.

[11] Jack Hopkins, Mart Bakler, and Akbir Khan. Factorio learning environment. *arXiv preprint arXiv:2503.09617*, 2025.

[12] Lanxiang Hu, Mingjia Huo, Yuxuan Zhang, Haoyang Yu, Eric P. Xing, Ion Stoica, Tajana Rosing, Haojian Jin, and Hao Zhang. lmgame-bench: How good are LLMs at playing games? In *The Fourteenth International Conference on Learning Representations*, 2026.

[13] Lanxiang Hu, Qiyu Li, Anze Xie, Nan Jiang, Ion Stoica, Haojian Jin, and Hao Zhang. Gamearena: Evaluating llm reasoning through live computer games. In *The Thirteenth International Conference on Learning Representations*, 2025.

[14] Frederikus Hudi, Genta Indra Winata, Ruochen Zhang, and Alham Fikri Aji. Textgames: Learning to self-play text-based puzzle games via language model reasoning. *arXiv preprint arXiv:2502.18431*, 2025.

[15] Seth Karten, Andy Luu Nguyen, and Chi Jin. Pok\'echamp: an expert-level minimax language agent. *arXiv preprint arXiv:2503.04094*, 2025.

[16] Kunchang Li, Yali Wang, Yinan He, Yizhuo Li, Yi Wang, Yi Liu, Zun Wang, Jilan Xu, Guo Chen, Ping Luo, et al. Mvbench: A comprehensive multi-modal video understanding benchmark. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 22195–22206, 2024.

[17] Marlos C Machado, Marc G Bellemare, Erik Talvitie, Joel Veness, Matthew Hausknecht, and Michael Bowling. Revisiting the arcade learning environment: Evaluation protocols and open problems for general agents. *Journal of Artificial Intelligence Research*, 61:523–562, 2018.

[18] Volodymyr Mnih, Koray Kavukcuoglu, David Silver, Andrei A Rusu, Joel Veness, Marc G Bellemare, Alex Graves, Martin Riedmiller, Andreas K Fidjeland, Georg Ostrovski, et al. Human-level control through deep reinforcement learning. *nature*, 518(7540):529–533, 2015.

[19] Muhammad Umair Nasir, Steven James, and Julian Togelius. Gametraversalbenchmark: Evaluating planning abilities of large language models through traversing 2d game maps. In *The Thirty-eight Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2024.

[20] Munan Ning, Bin Zhu, Yujia Xie, Bin Lin, Jiaxi Cui, Lu Yuan, Dongdong Chen, and Li Yuan. Video-bench: A comprehensive benchmark and toolkit for evaluating video-based large language models. *arXiv preprint arXiv:2311.16103*, 2023.

[21] Davide Paglieri, Bartłomiej Cupiał, Samuel Coward, Ulyana Piterbarg, Maciej Wolczyk, Akbir Khan, Eduardo Pignatelli, Łukasz Kuciński, Lerrel Pinto, Rob Fergus, Jakob Nicolaus Foerster, Jack Parker-Holder, and Tim Rocktäschel. BALROG: Benchmarking agentic LLM and VLM reasoning on games. In *The Thirteenth International Conference on Learning Representations*, 2025.

[22] Viorica Patraucean, Lucas Smaira, Ankush Gupta, Adria Recasens, Larisa Markeeva, Dylan Banarse, Skanda Koppula, Mateusz Malinowski, Yi Yang, Carl Doersch, et al. Perception test: A diagnostic benchmark for multimodal video models. *Advances in Neural Information Processing Systems*, 36:42748–42761, 2023.

[23] Dan Qiao, Chenfei Wu, Yaobo Liang, Juntao Li, and Nan Duan. Gameeval: Evaluating llms on conversational games. *arXiv preprint arXiv:2308.10032*, 2023.

[24] Anian Ruoss, Fabio Pardo, Harris Chan, Bonnie Li, Volodymyr Mnih, and Tim Genewein. LMAct: A benchmark for in-context imitation learning with long multimodal demonstrations. In *Forty-second International Conference on Machine Learning*, 2025.

[25] Mohammadreza Reza Salehi, Jae Sung Park, Aditya Kusupati, Ranjay Krishna, Yejin Choi, Hanna Hajishirzi, and Ali Farhadi. Actionatlas: A videoqa benchmark for domain-specialized action recognition. *Advances in Neural Information Processing Systems*, 37:137372–137402, 2024.

[26] Weihao Tan, Wentao Zhang, Xinrun Xu, Haochong Xia, Ziluo Ding, Boyu Li, Bohan Zhou, Junpeng Yue, Jiechuan Jiang, Yewen Li, Ruyi An, Molei Qin, Chuqiao Zong, Longtao Zheng, YuJie Wu, Xiaoqiang Chai, Yifei Bi, Tianbao Xie, Pengjie Gu, Xiyun Li, Ceyao Zhang, Long Tian, Chaojie Wang, Xinrun Wang, Börje F. Karlsson, Bo An, Shuicheng YAN, and Zongqing Lu. Cradle: Empowering foundation agents towards general computer control. In *Forty-second International Conference on Machine Learning*, 2025.

[27] Mark Towers, Ariel Kwiatkowski, Jordan Terry, John U Balis, Gianluca De Cola, Tristan Deleu, Manuel Goulão, Andreas Kallinteris, Markus Krimmel, Arjun KG, et al. Gymnasium: A standard interface for reinforcement learning environments. *arXiv preprint arXiv:2407.17032*, 2024.

[28] Nicholas R Waytowich, Devin White, MD Sunbeam, and Vinicius G Goecks. Atari-gpt: Benchmarking multimodal large language models as low-level policies in atari games. *arXiv preprint arXiv:2408.15950*, 2024.

[29] Junlin Xie, Ruifei Zhang, Zhihong Chen, Xiang Wan, and Guanbin Li. Whodunitbench: Evaluating large multimodal agents via murder mystery games. In *The Thirty-eight Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2024.

[30] Rui Yang, Hanyang Chen, Junyu Zhang, Mark Zhao, Cheng Qian, Kangrui Wang, Qineng Wang, Teja Venkat Koripella, Marziyeh Movahedi, Manling Li, et al. Embodiedbench: Comprehensive benchmarking multi-modal large language models for vision-driven embodied agents. In *Forty-second International Conference on Machine Learning*, 2025.

[31] Alex L Zhang, Thomas L Griffiths, Karthik R Narasimhan, and Ofir Press. Videogamebench: Can vision-language models complete popular video games? *arXiv preprint arXiv:2505.18134*, 2025.

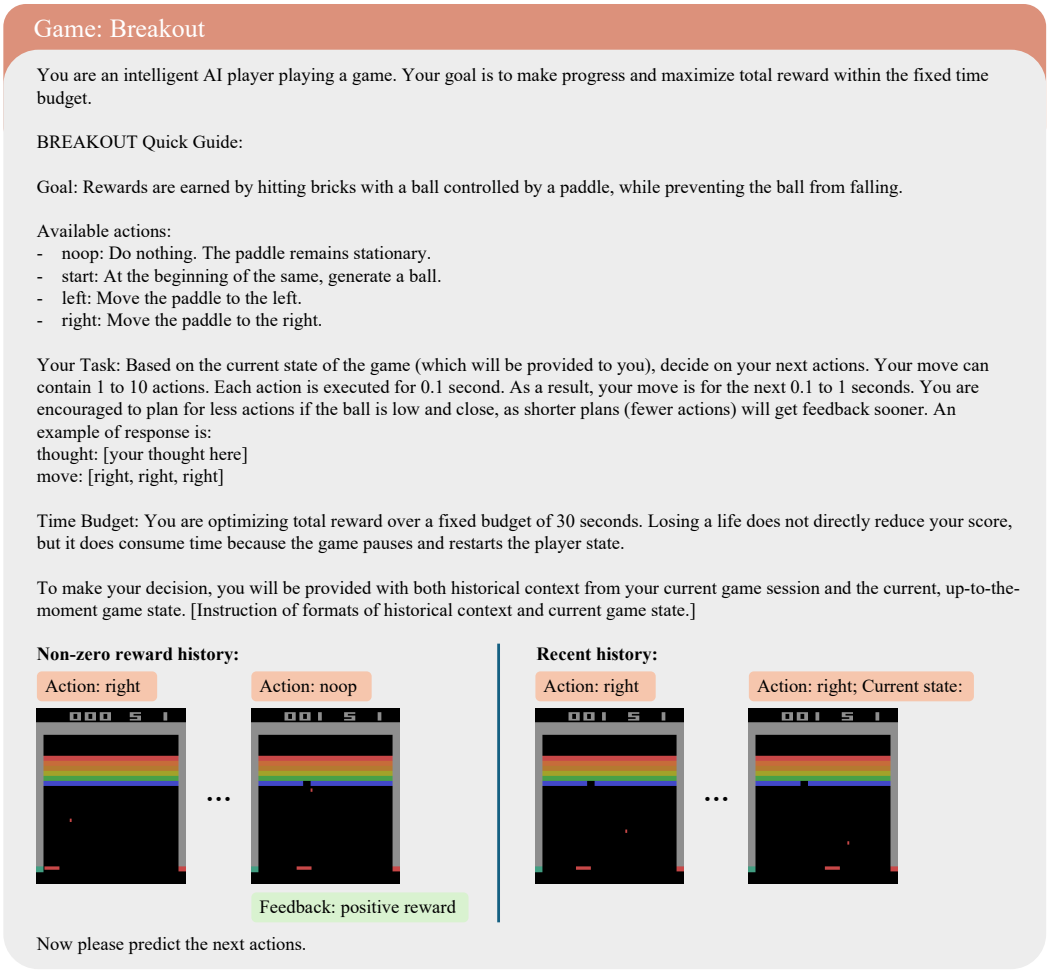


Figure 2: Example of game prompt.

A List of Games

Game Name	Adjusted #lives	Non-Zero Scoring Criteria
Breakout	5	-1.0: Player misses the ball 1.0: Player breaks a brick (equivalent to catch the ball)

B Statistics

In Figure X we show the models' performance. Here we list the other benchmarking statistics, including the length of game-play, number of tokens, and cost.

Table 3: Order-of-magnitude benchmark costs (illustrative), from estimated queries and average tokens per query.

Benchmark	Est. q	Model	Avg. in	Avg. out	Est. $
SWE-bench	2,294	gpt-5.4	100k	20k	$830
SWE-bench	2,294	claude-sonnet-4-6	100k	20k	$1,380
SWE-bench	2,294	gemini-3-pro	100k	20k	$750
AtariBench		gpt-5.4			
AtariBench		claude-sonnet-4-6			
AtariBench		gemini-3-pro			

Table 4: Estimated *per-step* costs (USD) from cached logs for two games under four harness settings.

Model	Game	No harness	Harness (perc)	Harness (mem)	Harness (both)
claude-3-7-sonnet-latest	candy_crush	0.011	0.021	0.047	0.048
claude-3-7-sonnet-latest	twenty_forty_eight	0.015	0.026	0.029	0.031
gemini-2.5-pro	candy_crush	0.005	0.009	0.019	0.020
gemini-2.5-pro	twenty_forty_eight	0.016	0.019	0.020	0.021
gpt-4.1	candy_crush	0.008	0.014	0.029	0.025
gpt-4.1	twenty_forty_eight	0.008	0.014	0.016	0.016
o1	candy_crush	0.055	0.105	0.212	0.218
o1	twenty_forty_eight	0.054	0.102	0.114	0.121

C Prompt

D Video input and image input

E Cost Analysis

Agentic evaluations are token-billed; major providers charge per input/output token, so longer contexts and extra tool calls increase spend.[1] Running AtariBench remains more controllable than full-scale software-agent benchmarks (e.g., SWE-bench) under comparable model sets and query counts. Tables 3–4 summarize order-of-magnitude totals and per-step estimates from cached logs. Overall, *with-harness* costs exceed *no-harness* due to (i) longer contexts from appended trajectories and (ii) an additional reflection call in the memory module; optional perception may add a vision-to-text call.

F Context Length Impact

To evaluate the impact of historical context on the models' performance, we compared the best scores achieved by Gemini models using two different lengths of context history: 3 clips and 30 clips, on five games. The results, as detailed in the Table 5, show that providing a longer context history of 30 clips generally leads to improved performance across most games. For instance, the average score in Assault increased significantly from 7.0 to 10.8, and in Pacman from 31.7 to 38.3. While the improvements in Breakout and Tennis were more modest, the overall trend indicates that most games benefit from a more extensive history context, which likely allows the model to make more informed, strategic decisions.

Table 5: Comparison of model performance with short (3 clips) and long (30 clips) context histories.

#history_clips	Assault	Breakout	Pacman	Tennis
3	7.0	6.2	31.7	-2.7
30	**10.8**	**6.4**	**38.3**	**-2.5**

[1] `https://openai.com/api/pricing`; `https://ai.google.dev/gemini-api/docs/billing`; `https://www.anthropic.com/news/1m-context`